SOA Actuarial Exam P – Complete Formula Reference Chart

Coverage: Phase 1 (General Probability), Phase 2 (Univariate Random Variables), and Phase 3 (Multivariate Random Variables)

Phase 1: General Probability

1. Set Operations & Event Algebra

- **Union ($A \cup B$):** A occurs, or B occurs, or both occur.
- **Intersection ($A \cap B$):** Both A and B occur simultaneously.
- **Complement (A^c or A'):** Event A does not occur.
- **Set Difference ($A \setminus B = A \cap B^c$):** Event A occurs, but event B does not occur.
- **De Morgan's Laws:**

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

- **Distributive Laws:**

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

- **Event Partitioning:**

$$A = (A \cap B) \cup (A \cap B^c) \quad \text{where } (A \cap B) \cap (A \cap B^c) = \varnothing$$

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2. Kolmogorov's Axioms & Basic Probability Rules

- **Axiom 1 (Non-negativity):** $P(A) \geq 0$ for all $A \subseteq S$.
- **Axiom 2 (Normalization):** $P(S) = 1$.
- **Axiom 3 (Countable Additivity):** For mutually exclusive events $A_1, A_2, \ldots$:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

- **Empty Set:** $P(\varnothing) = 0$
- **Complement Rule:** $P(A^c) = 1 - P(A)$
- **Boundedness:** $0 \leq P(A) \leq 1$
- **Difference Rule:** If $A \subseteq B$, then $P(B \setminus A) = P(B \cap A^c) = P(B) - P(A)$.
- **Monotonicity:** If $A \subseteq B$, then $P(A) \leq P(B)$.
- **Law of Total Probability for Sets:** $P(A) = P(A \cap B) + P(A \cap B^c)$

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3. Probability Inequalities & Bounds

- **Bonferroni's Inequality (Two Events):**

$$P(A \cap B) \geq P(A) + P(B) - 1$$

- **Bonferroni's Inequality (n Events):**

$$P\left(\bigcap_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} P(A_i) - (n-1) = 1 - \sum_{i=1}^{n} P(A_i^c)$$

- **Intersection Bounds:**

$$\max(0, P(A) + P(B) - 1) \leq P(A \cap B) \leq \min(P(A), P(B))$$

- **Boole's Inequality:**

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$$

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4. Combinatorics & Enumeration Techniques

- **Multiplication Principle:** $\text{Total Outcomes} = n_1 \times n_2 \times \cdots \times n_k$
- **Addition Principle:** $\text{Total Options} = n_1 + n_2 + \cdots + n_m$ (for disjoint sets)
- **Permutation Formula (Order Matters):**

$$P(n, r) = {}_nP_r = \frac{n!}{(n-r)!}$$

- **Permutations with Indistinguishable Objects:**

$$\text{Distinct Arrangements} = \frac{n!}{n_1!\, n_2!\, \ldots\, n_k!} \quad \text{where } \sum n_i = n$$

- **Combination Formula (Order Does NOT Matter):**

$$C(n, r) = \binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

- **Combinatorial Identities:**

$$\binom{n}{r} = \binom{n}{n-r}, \quad \binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$$

- **Stars & Bars (Distributing n identical items into k distinct bins):**
  - *Bins can be empty ($x_i \geq 0$):* $\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$
  - *Bins cannot be empty ($x_i \geq 1$):* $\binom{n-1}{k-1}$
- **Proximity Rules:**
  - *Block Method (Must sit together):* $(\text{Arrangements of Super-Items}) \times (\text{Internal Arrangements inside Block})$
  - *Gap Method (No two adjacent):* $m! \times \binom{m+1}{r} \times r!$ (m unrestricted, r restricted)
- **Grid / Lattice Paths from $(0, 0)$ to (a, b):**

$$\text{Total Paths} = \binom{a+b}{a} = \binom{a+b}{b} = \frac{(a+b)!}{a!\, b!}$$

$$\text{Paths via Waypoint } (x_1, y_1) = \binom{x_1 + y_1}{x_1} \times \binom{(a - x_1) + (b - y_1)}{a - x_1}$$

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5. Addition Rules & Inclusion-Exclusion

- **Mutually Exclusive (Disjoint) Events:** $A \cap B = \varnothing \implies P(A \cap B) = 0$

$$P(A \cup B) = P(A) + P(B)$$

- **General Addition Rule (Two Events):**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **Inclusion-Exclusion Principle (Three Events):**

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

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6. Independence & Conditional Probability

- **Definition of Independent Events:**

$$P(A \cap B) = P(A) \cdot P(B) \iff P(A \mid B) = P(A) \iff P(B \mid A) = P(B)$$

- **Independence of Complements:** If A and B are independent, so are A and B^c, A^c and B, and A^c and B^c.
- **"At Least One" Independent Event:**

$$P\left(\bigcup_{i=1}^{n} A_i\right) = 1 - \prod_{i=1}^{n} (1 - P(A_i))$$

- **Conditional Probability:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad \text{for } P(B) > 0$$

- **Multiplication Rule:**

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

$$P(A \cap B \cap C) = P(A) \cdot P(B \mid A) \cdot P(C \mid A \cap B)$$

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7. Law of Total Probability & Bayes' Theorem

- **Law of Total Probability:** For a partition $B_1, B_2, \ldots, B_k$ of sample space S:

$$P(A) = \sum_{i=1}^{k} P(A \cap B_i) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i)$$

- **Bayes' Theorem:**

$$P(B_j \mid A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{k} P(A \mid B_i)P(B_i)}$$

- **Odds & Posterior Odds:**

$$\text{Odds}(B) = \frac{P(B)}{P(B^c)}, \quad \text{Posterior Odds } \frac{P(B \mid A)}{P(B^c \mid A)} = \frac{P(B)}{P(B^c)} \times \frac{P(A \mid B)}{P(A \mid B^c)}$$

Phase 2: Univariate Random Variables

1. Fundamental Probability Functions & Properties

- **Probability Mass Function (PMF - Discrete):** $p_X(x) = P(X = x)$

$$\sum_x p_X(x) = 1, \quad p_X(x) \geq 0$$

- **Probability Density Function (PDF - Continuous):** $f_X(x)$

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1, \quad f_X(x) \geq 0, \quad P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

- **Cumulative Distribution Function (CDF):** $F_X(x) = P(X \leq x)$
  - *Discrete:* $F_X(x) = \sum_{k \leq x} p_X(k)$
  - *Continuous:* $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt \implies f_X(x) = \frac{d}{dx} F_X(x)$
  - *Properties:* Non-decreasing, $F(-\infty) = 0$, $F(\infty) = 1$, Right-continuous ($P(a < X \leq b) = F(b) - F(a)$).

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2. Expectation, Moments, and Measures of Location/Dispersion

- **Expected Value (Mean μ):**

$$\mathbb{E}[X] = \begin{cases} \sum_x x \cdot p_X(x) & \text{Discrete} \\ \int_{-\infty}^{\infty} x \cdot f_X(x)\,dx & \text{Continuous} \end{cases}$$

- **Tail Formula for Expectation (Non-negative $X \geq 0$):**

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} P(X > k) \quad (\text{Discrete}), \quad \mathbb{E}[X] = \int_0^{\infty} (1 - F_X(x))\,dx \quad (\text{Continuous})$$

- **Expected Value of a Function $g(X)$ (LOTUS):**

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x) p_X(x) & \text{Discrete} \\ \int_{-\infty}^{\infty} g(x) f_X(x)\,dx & \text{Continuous} \end{cases}$$

- **Linearity of Expectation:** $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$
- **Variance (σ^2):**

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

- **Standard Deviation:** $\sigma_X = \sqrt{\text{Var}(X)}$
- **Coefficient of Variation:** $CV = \frac{\sigma_X}{\mu_X}$

- **Raw Moment (k-th):** $\mu'_k = \mathbb{E}[X^k]$
- **Central Moment (k-th):** $\mu_k = \mathbb{E}[(X-\mu)^k]$
- **Percentile / Quantile (p-th percentile x_p):** $F_X(x_p) = p \implies x_p = F_X^{-1}(p)$
- **Median (m):** $F_X(m) = 0.50$
- **Mode:** The value x that maximizes $p_X(x)$ or $f_X(x)$.

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3. Discrete Parametric Distributions

Distribution	PMF: $P(X=x)$	Mean $\mathbb{E}[X]$	Variance $\mathrm{Var}(X)$	Notes / MGF
Bernoulli(p)	$p^x(1-p)^{1-x},\ x \in \{0,1\}$	p	$p(1-p)$	$M(t) = (1-p) + pe^t$
Binomial(n,p)	$\binom{n}{x}p^x(1-p)^{n-x},\ x = 0,\dots,n$	np	$np(1-p)$	$M(t) = (1-p+pe^t)^n$
Geometric(p) *(Trial of 1st Success)*	$(1-p)^{x-1}p,\ x = 1,2,3,\dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	Memoryless: $P(X > s+t \mid X > s) = P(X > t)$
Geometric(p) *(Failures before 1st Success)*	$(1-p)^x p,\ x = 0,1,2,\dots$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$	Alternative SOA parametrization
Negative Binomial(r,p)	$\binom{x-1}{r-1}p^r(1-p)^{x-r},\ x = r, r+1,\dots$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	Trial of r-th success
Hypergeometric(N,K,n)	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}},\ x = 0,\dots,n$	$n\frac{K}{N}$	$n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}$	Sampling without replacement
Poisson(λ)	$\frac{e^{-\lambda}\lambda^x}{x!},\ x = 0,1,2,\dots$	λ	λ	$M(t) = e^{\lambda(e^t-1)}$

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4. Continuous Parametric Distributions

Distribution	PDF $f(x)$ / Support	CDF $F(x)$	Mean $\mathbb{E}[X]$	Variance $\mathrm{Var}(X)$
Continuous Uniform(a,b)	$\frac{1}{b-a},\ x \in [a,b]$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(θ) *(Scale θ)*	$\frac{1}{\theta}e^{-x/\theta},\ x \geq 0$	$1-e^{-x/\theta}$	θ	θ^2
Exponential(λ) *(Rate λ)*	$\lambda e^{-\lambda x},\ x \geq 0$	$1-e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma(α,θ)	$\frac{x^{\alpha-1}e^{-x/\theta}}{\theta^\alpha\Gamma(\alpha)},\ x \geq 0$	Incomplete Gamma	$\alpha\theta$	$\alpha\theta^2$

Distribution	PDF $f(x)$ / Support	CDF $F(x)$	Mean $\mathbb{E}[X]$	Variance $\mathrm{Var}(X)$
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$\Phi\left(\frac{x-\mu}{\sigma}\right)$	μ	σ^2
Pareto(α, θ)	$\frac{\alpha\theta^\alpha}{(x+\theta)^{\alpha+1}},\ x \geq 0$	$1 - \left(\frac{\theta}{x+\theta}\right)^\alpha$	$\frac{\theta}{\alpha-1}$ $(\alpha > 1)$	$\frac{\alpha\theta^2}{(\alpha-1)^2(\alpha-2)}$ $(\alpha > 2)$

- **Exponential Memoryless Property:** $P(X > s+t \mid X > s) = P(X > t) = e^{-t/\theta}$
- **Standard Normal Transformation (Z-score):** $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$

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5. Insurance Policy Modifications & Loss/Payment Variables

Let X be the ground-up loss random variable ($X \geq 0$).

A. Ordinary Deductible (d)

- **Payment per loss variable (Y^L):**

$$Y^L = (X-d)_+ = \max(0, X-d) = \begin{cases} 0 & X \leq d \\ X-d & X > d \end{cases}$$

$$\mathbb{E}[Y^L] = \int_d^\infty (x-d) f_X(x)\,dx = \int_d^\infty (1 - F_X(x))\,dx$$

- **Payment per claim variable (Y^C):**

$$Y^C = (X-d \mid X > d)$$

$$\mathbb{E}[Y^C] = \frac{\mathbb{E}[Y^L]}{P(X>d)} = \frac{\mathbb{E}[Y^L]}{1 - F_X(d)}$$

B. Benefit Limit / Policy Maximum (u)

- **Coverage limited to maximum loss u:**

$$Y = \min(X, u) = \begin{cases} X & X < u \\ u & X \geq u \end{cases}$$

$$\mathbb{E}[\min(X,u)] = \int_0^u x f_X(x)\,dx + u(1 - F_X(u)) = \int_0^u (1 - F_X(x))\,dx$$

C. Deductible (d) AND Maximum Limit (u)

- **Maximum payment capped at $u-d$:**

$$Y^L = \begin{cases} 0 & X \leq d \\ X-d & d < X < u \\ u-d & X \geq u \end{cases}$$

$$\mathbb{E}[Y^L] = \int_d^u (1 - F_X(x))\,dx$$

D. Coinsurance (c) and Inflation (r)

- **With Coinsurance factor** $c \in (0,1]$**:** $Y^L = c \cdot (X-d)_+$
- **With Inflation rate** r **(Loss becomes** $X(1+r)$**):**

$$\mathbb{E}[\text{Payment with inflation}] = (1+r)\int_{d/(1+r)}^{\infty} \left(x - \frac{d}{1+r}\right) f_X(x)\,dx$$

Phase 3: Multivariate Random Variables

1. Joint, Marginal, and Conditional Distributions

A. Discrete Joint Distributions

- **Joint PMF:** $p_{X,Y}(x,y) = P(X=x, Y=y)$ where $\sum_x \sum_y p_{X,Y}(x,y) = 1$
- **Marginal PMFs:**

$$p_X(x) = \sum_y p_{X,Y}(x,y), \quad p_Y(y) = \sum_x p_{X,Y}(x,y)$$

- **Conditional PMF:**

$$p_{Y|X}(y \mid x) = P(Y=y \mid X=x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$$

B. Continuous Joint Distributions

- **Joint PDF:** $f_{X,Y}(x,y)$ where $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx\,dy = 1$
- **Joint CDF:**

$$F_{X,Y}(x,y) = P(X \le x, Y \le y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$$

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$$

- **Marginal PDFs:**

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

- **Conditional PDFs:**

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}, \quad f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

- **Joint Independence:** X and Y are independent if and only if:

$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y) \quad \text{for all } (x,y)$$

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2. Expectation, Covariance, and Correlation

A. Expectations of Joint Functions

$$\mathbb{E}[g(X,Y)] = \begin{cases} \sum_x \sum_y g(x,y) p_{X,Y}(x,y) & \text{Discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y)\, dx\, dy & \text{Continuous} \end{cases}$$

B. Conditional Expectation & Variance

- **Conditional Expectation:**

$$\mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x)\, dy$$

- **Law of Total Expectation (Double Expectation):**

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$$

- **Conditional Variance:**

$$\mathrm{Var}(Y \mid X = x) = \mathbb{E}[Y^2 \mid X = x] - (\mathbb{E}[Y \mid X = x])^2$$

- **Law of Total Variance (EVVE Rule):**

$$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$$

C. Covariance & Correlation Coefficient

- **Covariance Definition:**

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

(Note: If X and Y are independent, $\mathrm{Cov}(X,Y) = 0$. The converse is not generally true).

- **Covariance Properties:**

$$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$$

$$\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X,Y)$$

$$\mathrm{Cov}(X + Y, Z) = \mathrm{Cov}(X,Z) + \mathrm{Cov}(Y,Z)$$

- **Pearson Correlation Coefficient (ρ):**

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \quad \text{where } -1 \le \rho_{X,Y} \le 1$$

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3. Linear Combinations of Random Variables

Let $L = aX + bY + c$:

- **Expectation of Linear Combination:**

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

- **Variance of General Linear Combination:**

$$\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y)$$

- **Variance of Linear Combination (Independent X, Y):**

$$\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y)$$

- **General n-Variable Linear Combination $S = \sum_{i=1}^{n} a_i X_i$:**

$$\mathbb{E}[S] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i]$$

$$\mathrm{Var}(S) = \sum_{i=1}^{n} a_i^2 \mathrm{Var}(X_i) + 2 \sum_{1 \le i < j \le n} a_i a_j \mathrm{Cov}(X_i, X_j)$$

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4. Order Statistics

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed (i.i.d.) random variables with common PDF $f(x)$ and CDF $F(x)$. Arrange in ascending order: $Y_{(1)} \le Y_{(2)} \le \cdots \le Y_{(n)}$.

A. Maximum Order Statistic $Y_{(n)} = \max(X_1, \ldots, X_n)$

- **CDF:** $F_{Y_{(n)}}(y) = [F(y)]^n$
- **PDF:** $f_{Y_{(n)}}(y) = n[F(y)]^{n-1} f(y)$

B. Minimum Order Statistic $Y_{(1)} = \min(X_1, \ldots, X_n)$

- **CDF:** $F_{Y_{(1)}}(y) = 1 - [1 - F(y)]^n$
- **PDF:** $f_{Y_{(1)}}(y) = n[1 - F(y)]^{n-1} f(y)$

C. General k-th Order Statistic $Y_{(k)}$

- **PDF:**

$$f_{Y_{(k)}}(y) = \frac{n!}{(k-1)!(n-k)!} [F(y)]^{k-1} [1 - F(y)]^{n-k} f(y)$$

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5. Central Limit Theorem (CLT) & Normal Approximations

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with mean μ and variance $\sigma^2 < \infty$.

- **Sum Distribution ($S_n = \sum_{i=1}^{n} X_i$):**

$$\mathbb{E}[S_n] = n\mu, \quad \mathrm{Var}(S_n) = n\sigma^2$$

$$S_n \approx N(n\mu, n\sigma^2) \quad \text{as } n \to \infty$$

- **Sample Mean Distribution ($\bar{X}_n = \frac{1}{n} S_n$):**

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{as } n \to \infty$$

- **Standardized CLT Z-score:**

$$Z = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

- **Continuity Correction (Discrete to Continuous Normal):** For integer-valued S_n, approximate $P(a \leq S_n \leq b)$ using:

$$P(a - 0.5 \leq S_n \leq b + 0.5) \approx P\left(\frac{a - 0.5 - n\mu}{\sigma\sqrt{n}} \leq Z \leq \frac{b + 0.5 - n\mu}{\sigma\sqrt{n}}\right)$$